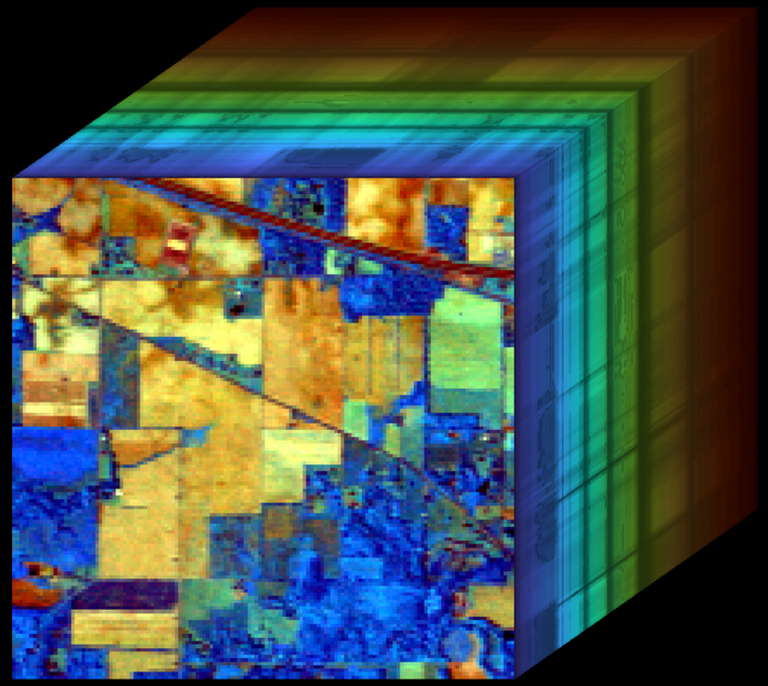
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PROJECT GHOST

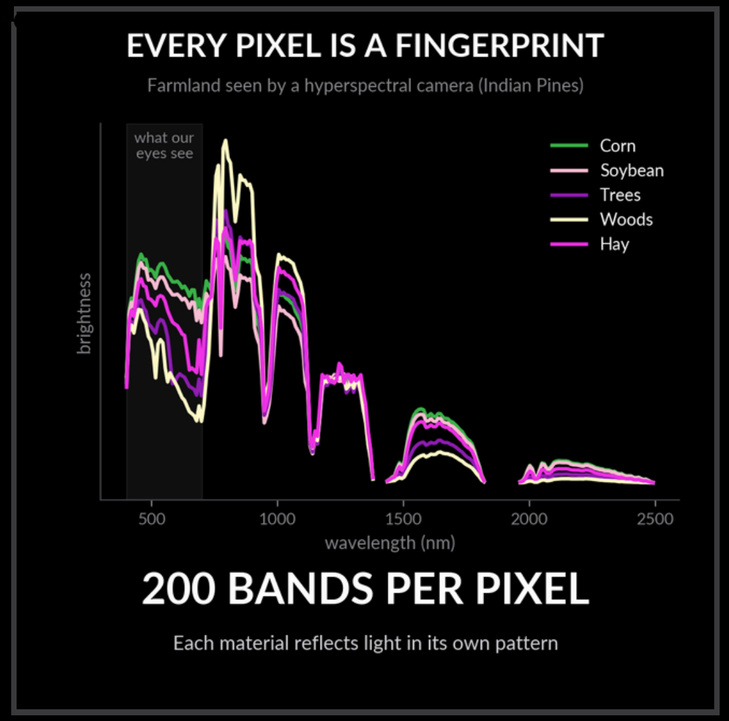
- **Problem Statement ID - SIH26209**
- **Problem Statement Title - Student Innovation-Space technology refers to the application of engineering principles to the design, development, manufacture, and operation of devices and systems for space travel and exploration.**
- **Theme - Space Technology**
- **PS Category - Software**
- **Team ID - 126234**
- **Team Name - Maneki_Neko**



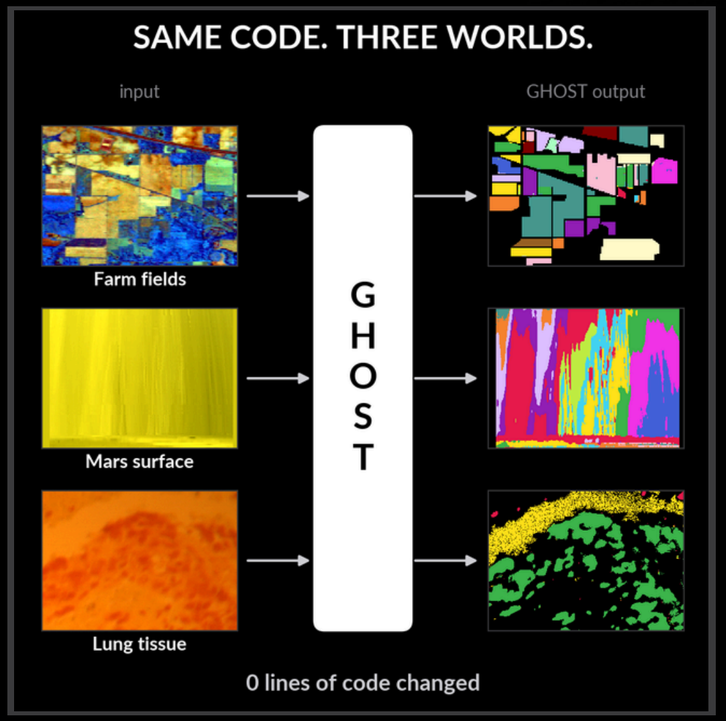
GHOST: GENERAL HYPERSPECTRAL OBSERVATION AND SEGMENTATION TOOLKIT

Maneki_Neko

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THE PROBLEM

A hyperspectral camera sees hundreds of colours per pixel, enough to identify materials by their chemistry.
But today's models must be rebuilt for every new sensor or scene: new code, new feature extraction (like PCA), new tuning.
Move one from farmland to Mars or to lung tissue and it fails. Every task means rewriting code.

PROPOSED SOLUTION

GHOST, a published Python package: point it at any hyperspectral cube, get a per-pixel material map.

HOW IT ADDRESSES THE PROBLEM

It reads bands and classes from the file. Same code on farm fields, the surface of Mars and inside your lungs.

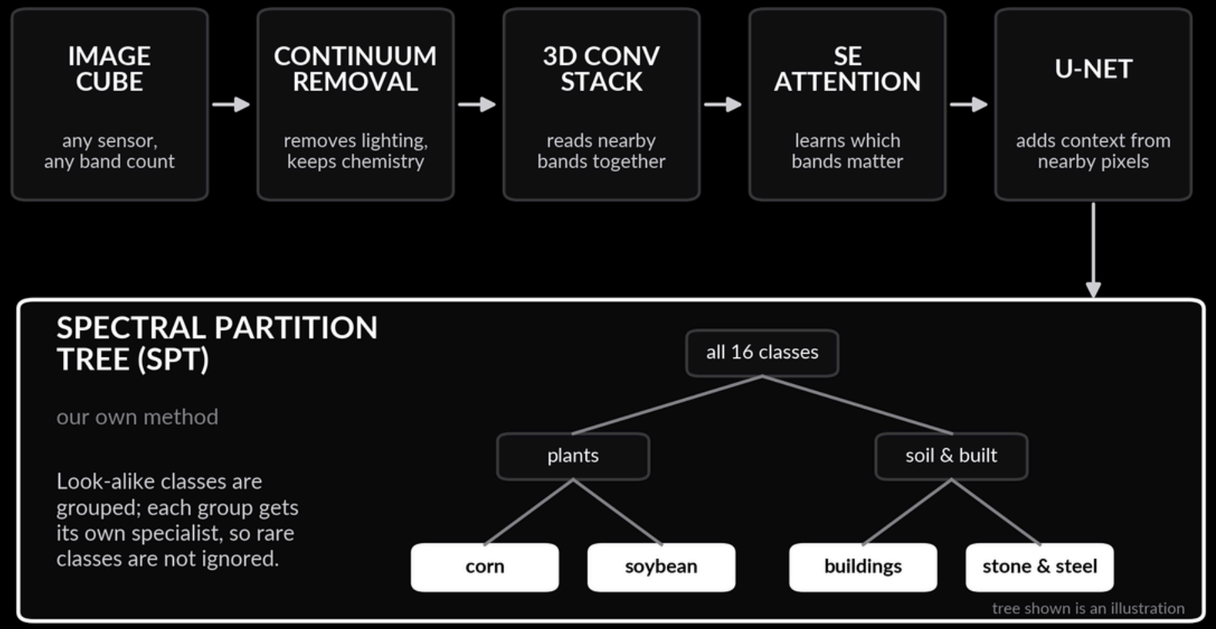
INNOVATION AND UNIQUENESS

Data, band-count and sensor agnostic: done.
Spectral-only context: v0.2.0, in production.

TECHNICAL APPROACH

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HOW GHOST READS AN IMAGE



DATASET	WHERE IT COMES FROM	BANDS	ACCURACY	mIoU
Indian Pines	farmland · airborne	200	**97.2%**	0.80
Salinas	farmland · airborne	204	**98.7%**	0.96
Pavia University	city · airborne	103	**97.5%**	0.95
LUSC	lung tissue · microscope	61	**99.4%**	0.93
Mars CRISM	Mars · orbiter	~400	**71.7%***	0.52

GHOST v0.1.7, published results, same code and settings on every dataset. * CRISM labels are sparse and noisy.

METHODOLOGY

Continuum removal strips out lighting and keeps the chemistry, so no PCA or manual preprocessing is needed.
3D convolutions and SE attention read neighbouring bands together and learn which ones matter. A U-Net adds context from nearby pixels.
Our own Spectral Partition Tree (SPT) groups look-alike classes and trains a specialist for each, so rare classes get learned instead of ignored. On Indian Pines it lifted per-class accuracy from 54% to 83%.

TECHNOLOGIES USED

Python · PyTorch · NumPy · SciPy · scikit-learn · Cython · rasterio · h5py

IT WORKS: THE PROOF

Same code and settings on five very different datasets: farmland, a city, lung tissue and the surface of Mars.
Above 97% accuracy on four of them, on a laptop GPU or a free Kaggle notebook.

▶ Full architecture: architecture.md on GitHub

Maneki_Neko

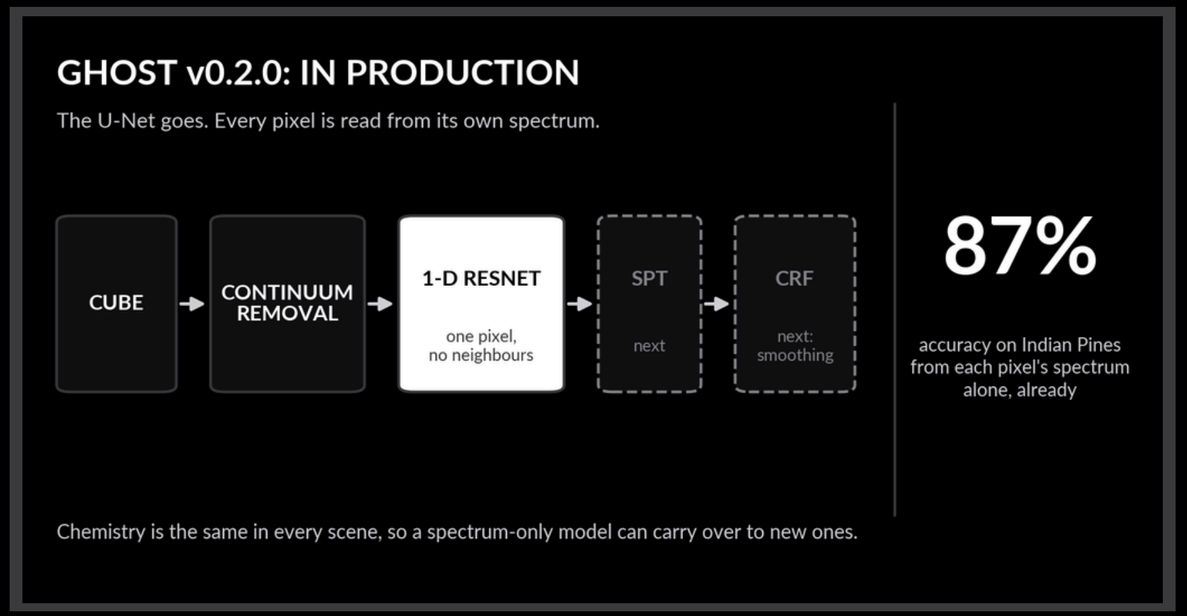
FEASIBILITY AND VIABILITY

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FEASIBILITY

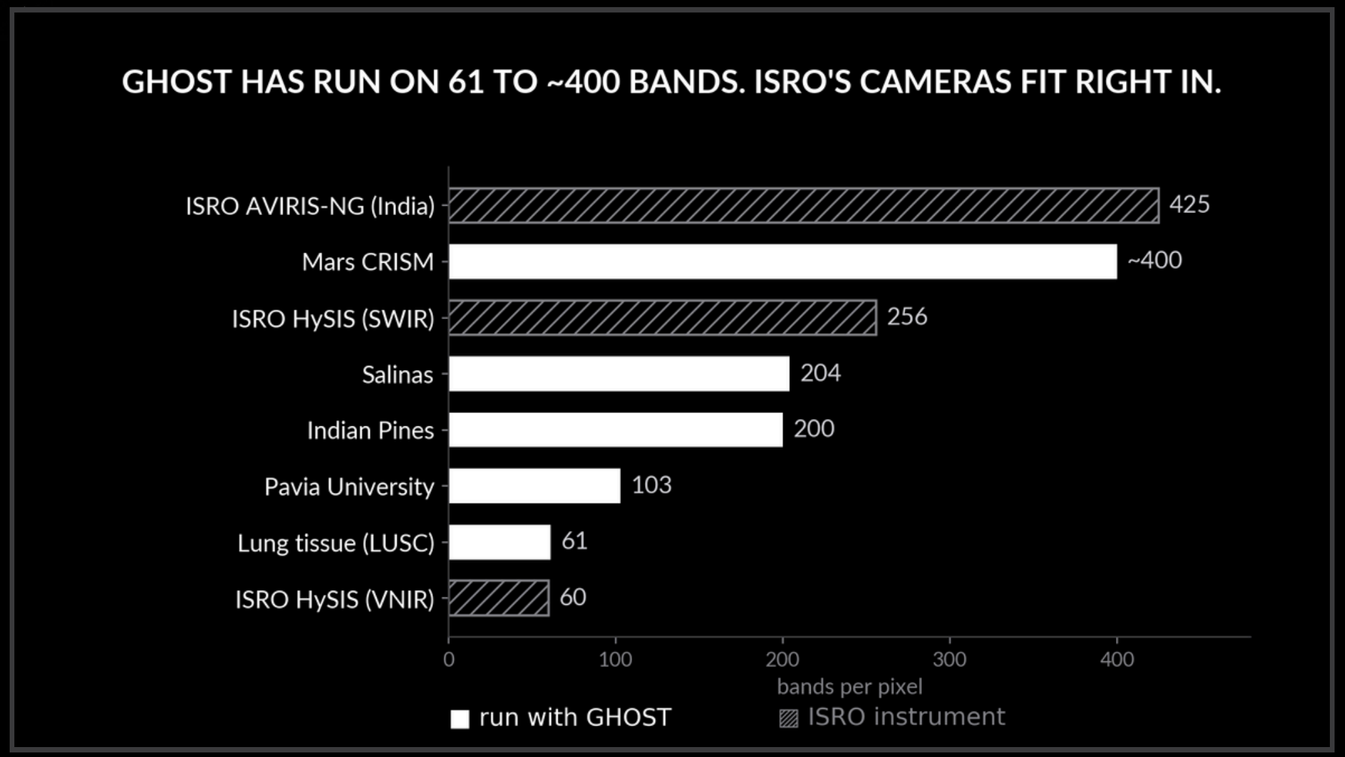
- Ready today: pip install ghost-hsi, then run. No setup, no code changes.
- Works on Windows, macOS and Linux, Python 3.9 to 3.12.
- Runs on free Google Colab and Kaggle notebooks, or a 6 GB laptop GPU.
- ISRO already flies hyperspectral cameras (HySIS, AVIRIS-NG, Chandrayaan-2 IIRS). Their images need no new code.



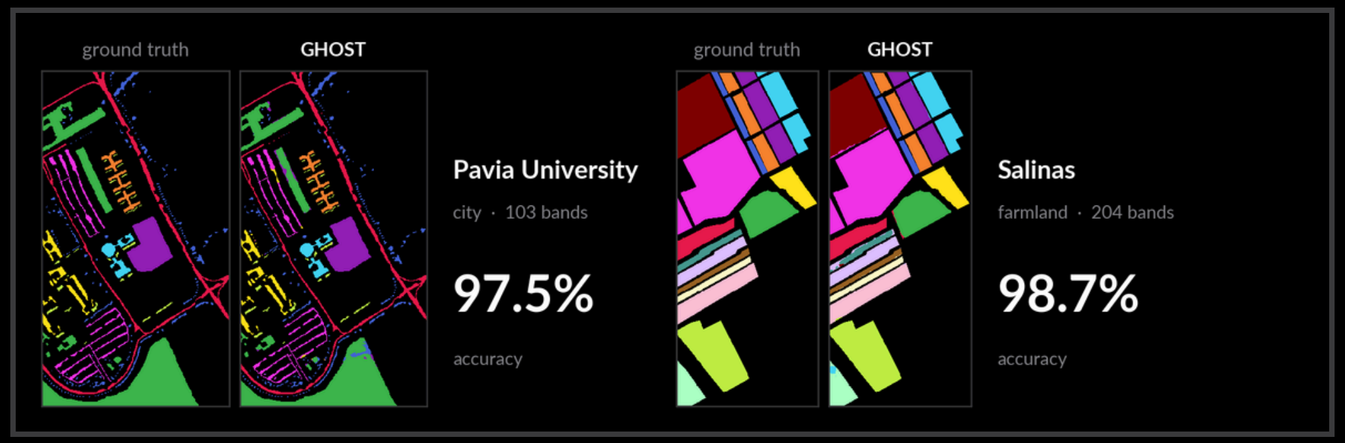
CHALLENGES → STRATEGIES

- NEIGHBOUR DEPENDENCE: v0.1.7 also learns from nearby pixels, so it partly learns where things sit in one scene. A model trained on one scene does not carry over to the next.
- WHY IT MATTERS: chemistry is the same everywhere; field layouts are not. → v0.2.0 reads each pixel's own spectrum only. It is already in production and scores 87% on Indian Pines without any neighbours.

IMPACT AND BENEFITS



Tested from 61 to ~400 bands. ISRO's HySIS and AVIRIS-NG images fall in this range.



IMPACT ON THE TARGET AUDIENCE

Researchers, ISRO scientists, students, farm and medical labs

- Earth: crops, cities, soil and minerals
- Space: Mars surface and asteroid data
- Medicine: lung cancer tissue slides
- Any new sensor: retrain, zero code changes

BENEFITS

- Zero code changes across sensors and domains
- Free and open source: pip install ghost-hsi
- Runs on a laptop GPU or free Colab
- Fits ISRO sensors: HySIS, AVIRIS-NG
- One tool: agriculture, geology, space, medicine

Any sensor. Any world.
Zero code changes.

RESEARCH AND REFERENCES

DATA AND SENSORS

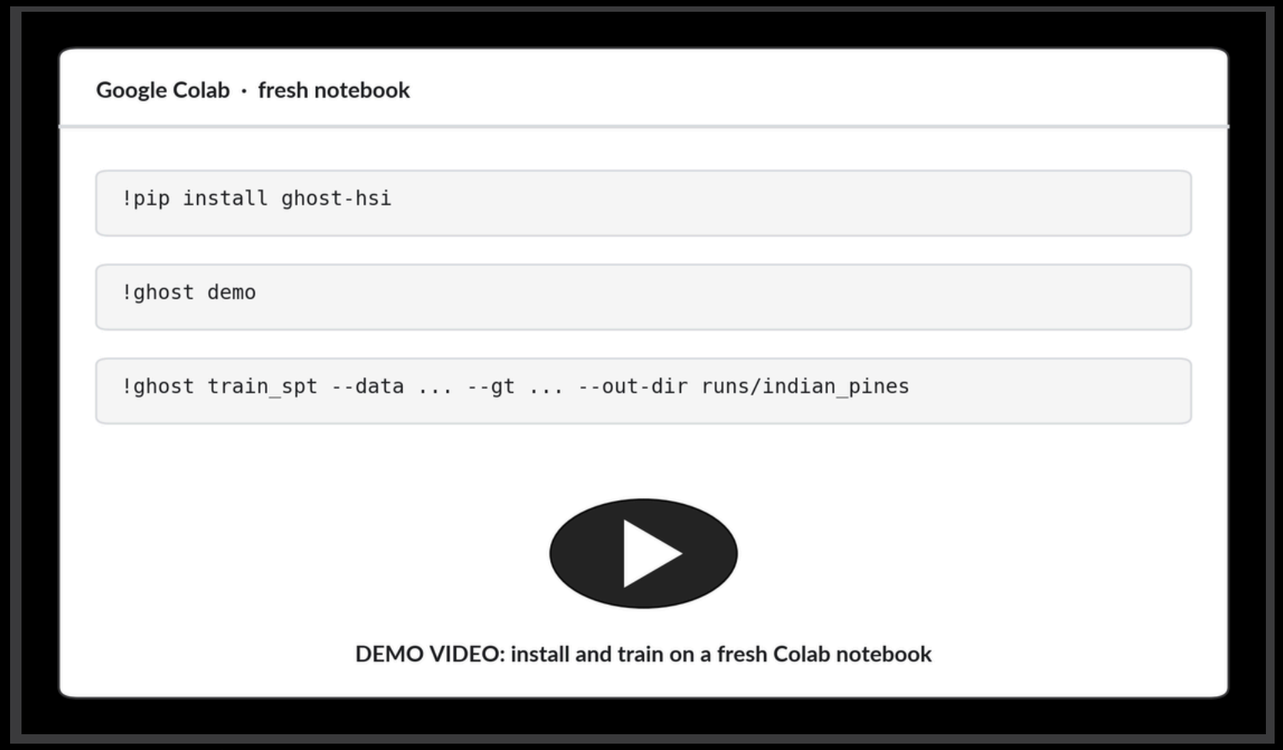
- HySIS: ISRO's hyperspectral imaging satellite (eoPortal)
- AVIRIS-NG India campaign, Phase 1 overview (SAC)
- Pavia and Salinas scenes (Computational Intelligence Group, UPV/EHU)
- HMI lung hyperspectral dataset (LUSC)
- Baumgardner et al.: Indian Pines AVIRIS data set (Purdue, 2015)
- Murchie et al.: CRISM on Mars Reconnaissance Orbiter (JGR, 2007)

THE METHOD

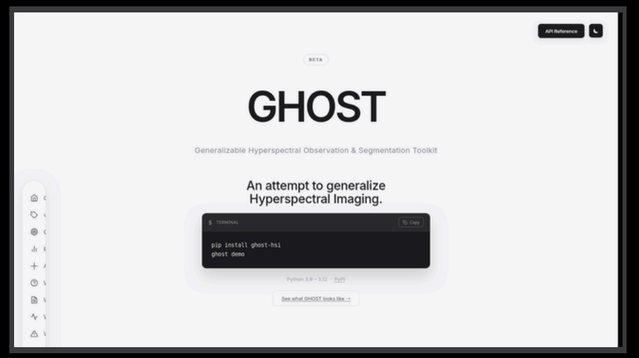
- Clark & Roush: continuum removal in reflectance spectroscopy (JGR, 1984)
- Hu et al.: squeeze-and-excitation networks (arXiv:1709.01507)
- Ronneberger et al.: U-Net (arXiv:1505.04597)
- He et al.: ResNet, the base of v0.2.0 (arXiv:1512.03385)
- Krähenbühl & Koltun: dense CRFs, planned for v0.2 (arXiv:1210.5644)

OURS

- Code: github.com/IshuIsAwake/GHOST
- Package: pypi.org/project/ghost-hsi (pip install ghost-hsi)
- Website: ishuisawake.github.io/GhostWEB



▶ DEMO VIDEO: from pip install to training, on a fresh Google Colab notebook



▶ Visit the website to know more: ishuisawake.github.io/GhostWEB